

SECURING **Xen** VIRTUALISATION



The concept of virtualisation evolved in order to achieve easy management and maintenance of the server infrastructure. While there are many production environments running on robust virtual servers, there are underlying security challenges that should be carefully addressed while designing virtualisation solutions. This article talks about a Xen virtualisation solution and also explains the measures to make it secure. Network administrators will find these techniques ideal to protect their virtual server infrastructure.

Simply put, a virtual server is an instance of an operating system, which shares system resources such as CPU time and memory to simulate a computer machine. At the heart of the operating system lies a component called HAL (hardware abstraction layer), which converts the CPU instruction set into meaningful operating system calls, which are further used by high level application layers. Virtualisation is a software layer that runs along with the HAL and operating system, to allow the creation of machine instances called guest systems. Guest systems that simulate a computer are capable of running any operating system such as Windows, Linux, etc. This becomes possible because each instance owns a configurable amount of CPU time and memory resources. Thus on a powerful physical machine, multiple varieties of guest operating systems can run

in parallel. This is similar to an operating system running several applications in multi-tasking mode; the difference being that, in this case, several operating systems run using a shared CPU model in a multi-threading mode.

It is important to note that, other than sharing just the CPU and memory resources, the peripherals such as the CD-ROM and USB ports are shared too. The most important component is the network interface card. Typically, in a physical server hosting virtual guests, two or more network cards are hosted, which are shared in a teaming mode, to provide redundancy if one interface fails. It is also important to note here that a virtual session manager is running in parallel to keep track of all guest instances, and hence consumes some amount of systems resources too.